

Sequence Analysis

Adding a time component to
Association Analysis

Sequence Analysis

- Adds a time component to Association Analysis
- Given that one **already** purchased item A, what is the probability that they will purchase item B in the future?

Customer ID	7/01/15	7/02/15	7/03/15	7/04/15
1001	hammer nails	nails		
1002	drill screws	saw	drill bits	
1003	hammer drill	screws drill bits	nails	

Sequence Analysis Applications

- Customer purchase history
 - What are they likely to buy next?
- Web analytics browsing activity
 - Where are they likely to click next?
- Event data generated by sensors/warranty claims
 - What part is likely to fail next?
- Genomic Sequences
 - What DNA Base (or group of bases) should we expect to find next?
- Typical Care Patterns
 - Given a diagnosis/procedure, what other events should we expect to follow?
- Use your imagination!

Sequence Analysis Inputs

- Transactional data
 - Customer ID
 - Sequence Component
 - Date
 - Sequence Counter 1,2,3,...

Some Considerations for Inputs

Consolidation window

- What consecutive visits/transactions should be considered a single transaction?
- Same hour? Same day? Same week?

Transaction window

- At what window of time do you want to stop considering the transactions for rules?
- Event A happens then Event B happens 1 week later? 1 month later? 1 year later? At what point do we ignore the potential association?

Quantifying Sequential Association Rules

For the rule Event A $\rightarrow$ Event B...

- **Support.** Of all the sequences considered, what proportion contain A before B.
- **Confidence.** Of all the sequences where A happened, in what proportion did B follow?
- **Pseudolift.** How much more likely are we to see B after A than we are to see B at random?

Quantifying Sequential Association Rules

For the rule Event A $\rightarrow$ Event B...

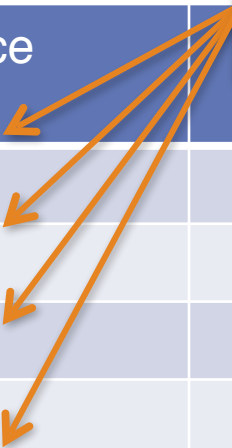
- *Support.*
- *Confidence.*
- *Pseudolift.*

All of these statistics will change depending on the transaction window and consolidation window!

Example

Machine ID	Fault Code Sequence	
1	{A} {D,F} {} {}	
2	{C,A} {D} {F} {}	
3	{D,C} {A} {} {B}	
4	{A} {C} {D,F} {A}	
5	{C} {} {A,D} {F}	

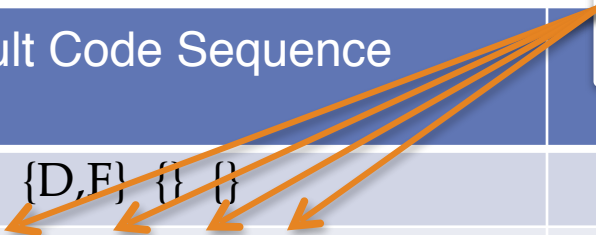
Fault codes reported
at each timestamp.



Example

Machine ID	Fault Code Sequence	
1	{A} {D,F} {} {}	
2	{C,A} {D} {F} {}	
3	{D,C} {A} {} {B}	
4	{A} {C} {D,F} {A}	
5	{C} {} {A,D} {F}	

Total timeline in data:
4 timestamps



Example (1a)

FOR RULE A → D

support = 3/5

confidence = 3/5

pseudolift = 3/5

Machine ID	Fault Code Sequence	Subsequence A → D?
1	{A} {D,F} {} {}	yes
2	{C,A} {D} {F} {}	yes
3	{D,C} {A} {} {B}	no
4	{A} {C} {D,F} {A}	yes
5	{C} {} {A,D} {F}	no

Consolidate time: 0 timestamps

Transaction window: All time

Example (1b)

FOR RULE A → D

support = 2/5

confidence = 2/5

pseudolift = 2/5

Machine ID	Fault Code Sequence	Subsequence A → D?
1	{A} {D,F} {} {}	yes
2	{C,A} {D} {F} {}	yes
3	{D,C} {A} {} {B}	no
4	{A} {C} {D,F} {A}	no
5	{C} {} {A,D} {F}	no

Consolidate time: 0 timestamps

Transaction window: 1 timestamps

Example (2a)

FOR RULE C- $\rightarrow$ A

support = 3/5

confidence = 3/4

pseudolift = 3/4

Machine ID	Fault Code Sequence	Subsequence C- $\rightarrow$ A?
1	{A} {D,F} {} {}	no
2	{C,A} {D} {F} {}	no
3	{D,C} {A} {} {B}	yes
4	{A} {C} {D,F} {A}	yes
5	{C} {} {A,D} {F}	yes

Consolidate time: 0 timestamps

Transaction window: All time

Example (2b)

FOR RULE $C \rightarrow A$

support = $2/5$

confidence = $2/4$

pseudolift = $2/4$

Machine ID	Fault Code Sequence	Subsequence $C \rightarrow A$?
1	{A} {D,F} {} {}	no
2	{C,A} {D} {F} {}	no
3	{D,C} {A} {} {B}	no
4	{A} {C} {D,F} {A}	yes
5	{C} {} {A,D} {F}	yes

Consolidate time: 1 timestamp

Transaction window: All time

Example (3a)

FOR RULE C-> F

support = 3/5

confidence = 3/4

pseudolift = $(3/4) \div (4/5) = 0.94$

Machine ID	Fault Code Sequence	Subsequence C->F?
1	{A} {D,F} {} {}	no
2	{C,A} {D} {F} {}	yes
3	{D,C} {A} {} {B}	no
4	{A} {C} {D,F} {A}	yes
5	{C} {} {A,D} {F}	yes

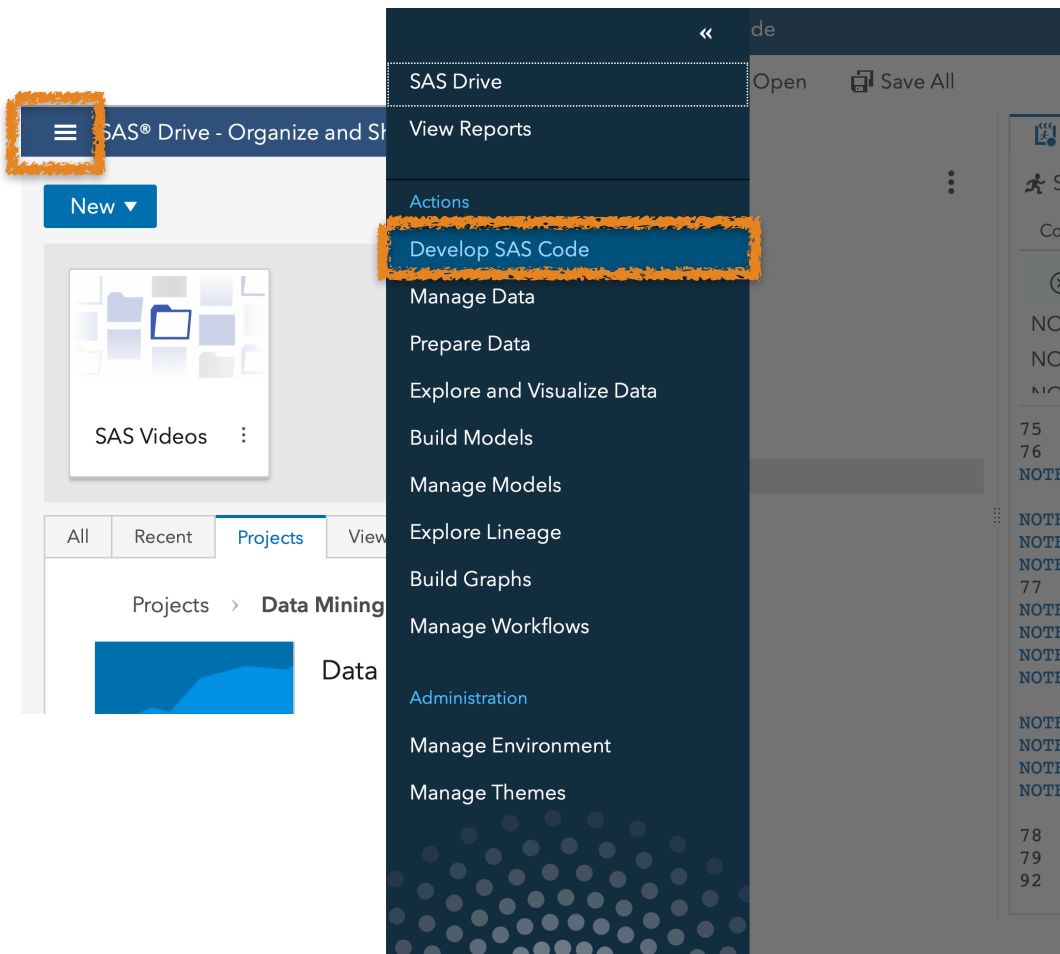
Consolidate time: 0 timestamps

Transaction window: All time

Lift and Pseudolift

- Values of lift and pseudolift greater than 1 indicate an association rule with some predictive power.
- In the previous examples, all lifts < 1 because the *consequents* of the rules happened frequently.

Path Analysis in Viya

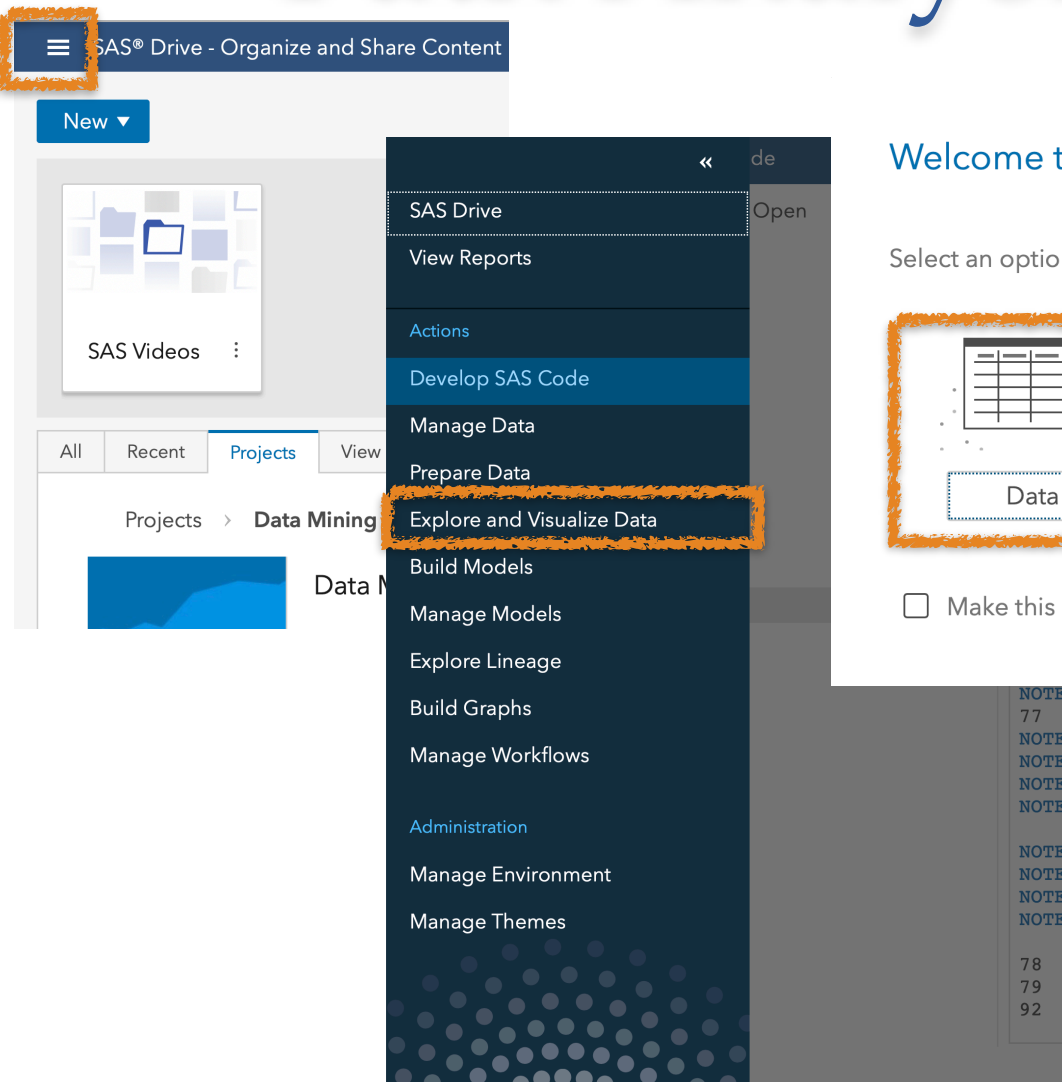


Submit Code:

```
cas;  
caslib _all_ assign;
```

You will repeat this step
EVERY time you use Viya
to load the Public library!

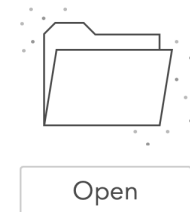
Path Analysis in Viya



The screenshot displays the SAS Drive interface. At the top, a dark blue header bar contains the text "SAS® Drive - Organize and Share Content" next to a hamburger menu icon. Below this, a "New" button is visible. The main area shows a "SAS Videos" section with a folder icon. A sidebar on the right lists various actions: "SAS Drive", "View Reports", "Actions", "Develop SAS Code", "Manage Data", "Prepare Data", "Explore and Visualize Data" (highlighted with an orange dashed border), "Build Models", "Manage Models", "Explore Lineage", "Build Graphs", "Manage Workflows", "Administration", "Manage Environment", and "Manage Themes". The "Explore and Visualize Data" option is also highlighted with an orange dashed border. The bottom of the sidebar shows a "Data Mining" section with a blue wave icon.

Welcome to SAS Visual Analytics

Select an option to get started:



☐ Make this selection the default

Dataset: public.FaultCodes

Data Setup

△ vehicleID	△ Code	⊛ sequence	⊛ numFaults
1024087	C9876	1	2
1024087	F7142	2	2
1006535	A1002	1	2
1006535	C5151	2	2
1025431	B2953	1	3
1025431	F1111	2	3
1025431	C9876	3	3
1014587	A1034	1	3
1014587	B2953	2	3
1014587	F5722	3	3

Visual Path Analysis

Objects



Data



Objects



Outline

Filter

▼ Controls

Button Bar

Drop-Down List

List

Slider

Text Input

▼ Analytics

Automated Analysis

Forecasting

Network Analysis

Path Analysis

Drag object
to chart area

Data Roles

Path Analysis - Code 1

▼ Event

Code

▼ Sequence order

sequence

▼ Transaction identifier

vehicleID

▼ Weight

Frequency



Options



Roles



Actions



Rules



Filters



Ranks

